Chapter 7

Perfect Competition and the Invisible Hand

ECON 003
UC Riverside · Summer 2026



Where We Left Off

Chapter 5 built the demand curve out of one buyer's problem.
Chapter 6 built the supply curve out of one seller's problem.
Today we put them in the same room.

WHAT BUYERS DO

Spend the next dollar where it buys the most utility. Stack those choices and you have demand.

WHAT SELLERS DO

Produce where price equals marginal cost. Stack those choices and you have supply.

WHAT NOBODY DOES

Coordinate any of it. No one adds the two sides up. And yet the market clears.

Today's question: when self-interested people trade, what does society end up with?

Today's Roadmap



1. Efficiency, measured



2. Across industries



3. Prices as signals



4. Equity vs efficiency

PART 1

Perfect Competition and Efficiency

One Sentence, from 1776



"It is not from the benevolence of the butcher, the brewer, or the baker, that we expect our dinner, but from their regard to their own interest."

Adam Smith, The Wealth of Nations, 1776

Self-interested ≠ selfish
Maximum well-being ≠ fairness

We are examining how efficient markets lead to the maximum level of well-being.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Two Numbers That Do All the Work

Reservation value — the most a buyer will pay, or the least a seller will accept, for one unit.

Buyer's surplus = reservation value − price · Seller's surplus = price − reservation value

FOR A BUYER



It is the value of the thing to you. Above it you walk away; below it you buy and pocket the difference.

FOR A SELLER



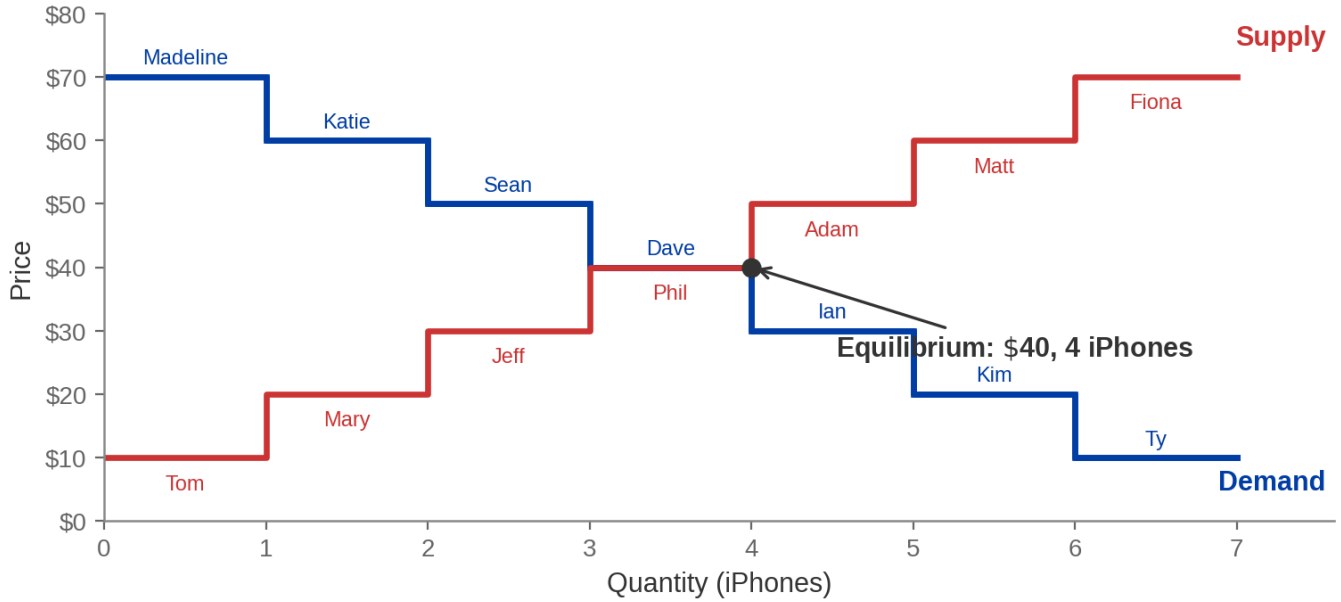
It is what the thing costs you to give up. Below it you keep it; above it you sell and pocket the difference.

Exhibit 7.1 — Seven Buyers, Seven Sellers

Buyer	Will pay	Cum. Q	Seller	Will accept	Cum. Q
Madeline	$70	1	Tom	$10	1
Katie	$60	2	Mary	$20	2
Sean	$50	3	Jeff	$30	3
Dave	$40	4	Phil	$40	4
Ian	$30	5	Adam	$50	5
Kim	$20	6	Matt	$60	6
Ty	$10	7	Fiona	$70	7

One market for one used phone each. Nobody trades more than once.

Exhibit 7.2 — The Same Table, Drawn



Every step is one person. Where the staircases cross is where the table balanced.

Consumer Surplus, at a Price of $40

Buyer	Will pay	Pays	Consumer surplus
Madeline	$70	$40	$30
Katie	$60	$40	$20
Sean	$50	$40	$10
Dave	$40	$40	$0
Ian, Kim, Ty	$30 and below	— no trade	$0
Total			$60

Consumer surplus is what buyers got for free: the gap between what a phone was worth to them and what they paid.

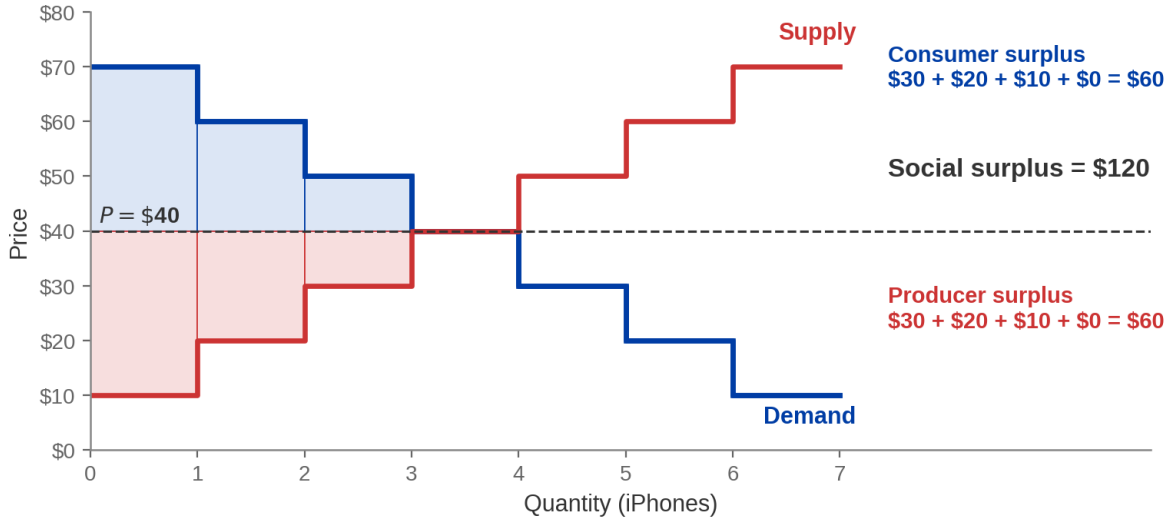
Producer Surplus, at the Same Price

Seller	Will accept	Receives	Producer surplus
Tom	$10	$40	$30
Mary	$20	$40	$20
Jeff	$30	$40	$10
Phil	$40	$40	$0
Adam, Matt, Fiona	$50 and above	— no trade	$0
Total			$60

Producer surplus is the mirror image: the gap between what a seller received and the least they would have accepted.

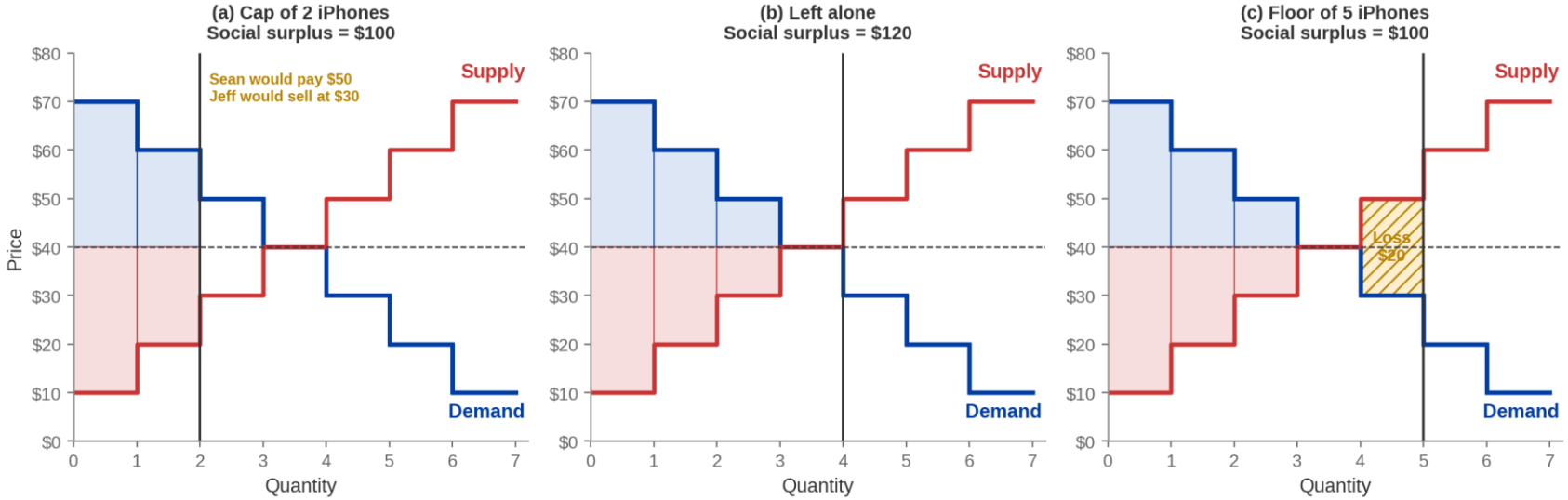
Social Surplus

Social surplus = consumer surplus + producer surplus



$60 + $60 = $120. That is the whole value this market created — now try to beat it.

Exhibit 7.3 — Cap It, Leave It, or Force It



Too few trades wastes $20. Too many wastes $20. Left alone, the market finds the top.

Pareto Efficiency

An allocation is Pareto efficient when nobody can be made better off without making somebody else worse off.

AT $40 AND FOUR PHONES

Try it. Move a phone to Ian and you must take it from someone who values it more. Change the price and one side gains exactly what the other loses. There is no free move left.

AT $20, SAY

Buyers do better and sellers do worse. That is not a Pareto improvement — it is a transfer. And with fewer sellers willing, fewer phones trade at all.

“No free moves left” is a much weaker claim than “this outcome is fair”.

What “Efficient” Does and Does Not License

IT DOES SAY

If the market is perfectly competitive and you intervene, you will end up with less total surplus than you started with. That is a real result and it has teeth.

IT DOES NOT SAY

That the outcome is fair, that everyone can afford the product, or that we should never intervene. Surplus counts dollars, not people.

So intervention cannot be justified on efficiency grounds here — but it may still be justified on other grounds.

Efficiency is one argument. It was never meant to be the only one.

PART 2

Across Industries

The Three Steps, One More Time

1. Find q where MR = MC · 2. Read ATC at that q · 3. Compare P with ATC

P > ATC



Economic profit. The industry is paying better than the alternatives, and outsiders can see it.

P < ATC



Economic loss. Resources here are earning less than they would elsewhere.

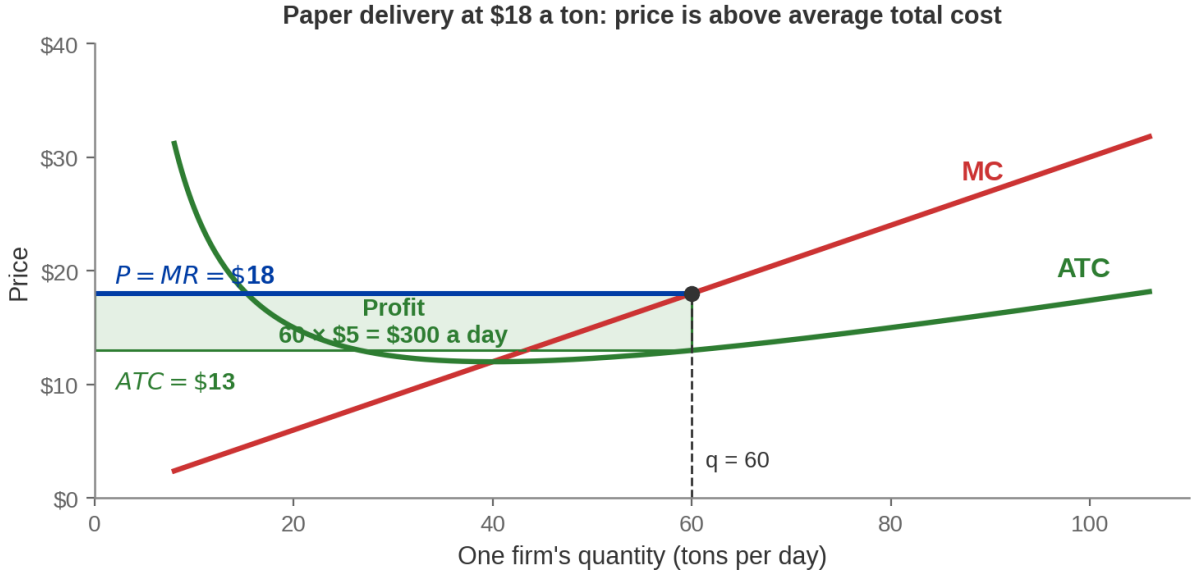
P = ATC



Break-even. Nothing pulls anyone in and nothing pushes anyone out.

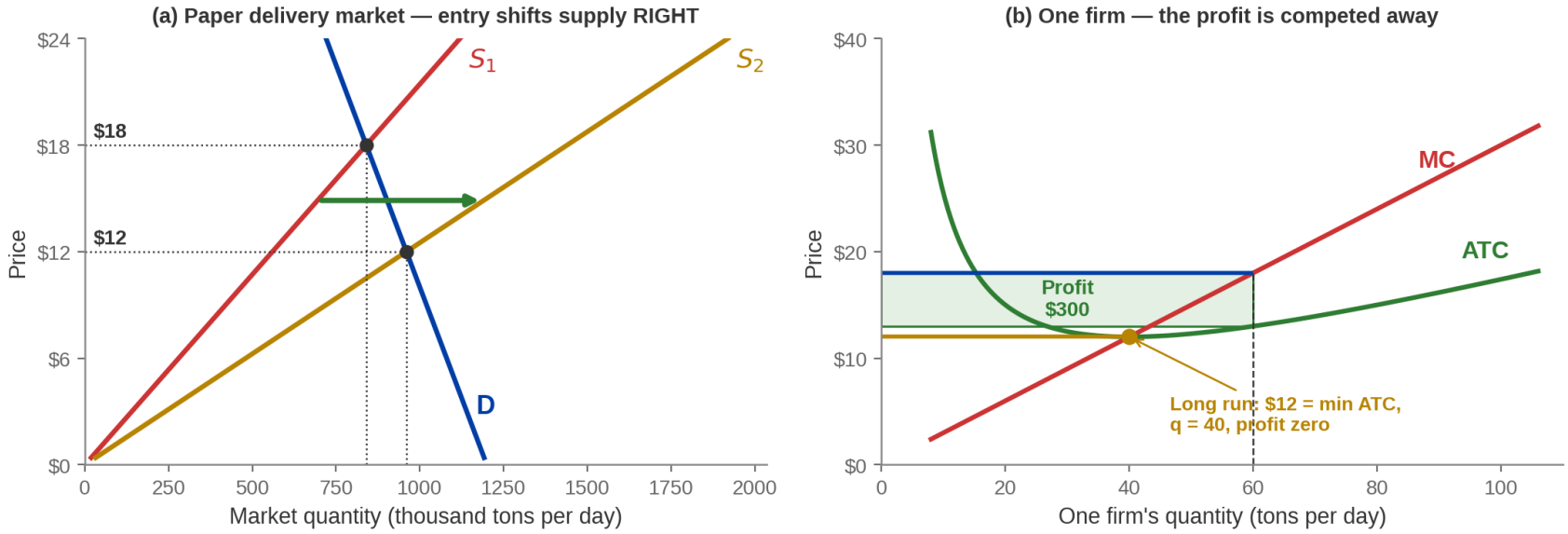
Economic profit is not a reward. It is a signal, and everyone can read it.

Exhibit 7.9 — Economic Profit in Paper Delivery



q = 60 tons a day · price $18 · ATC $13 · profit $300 a day, per firm

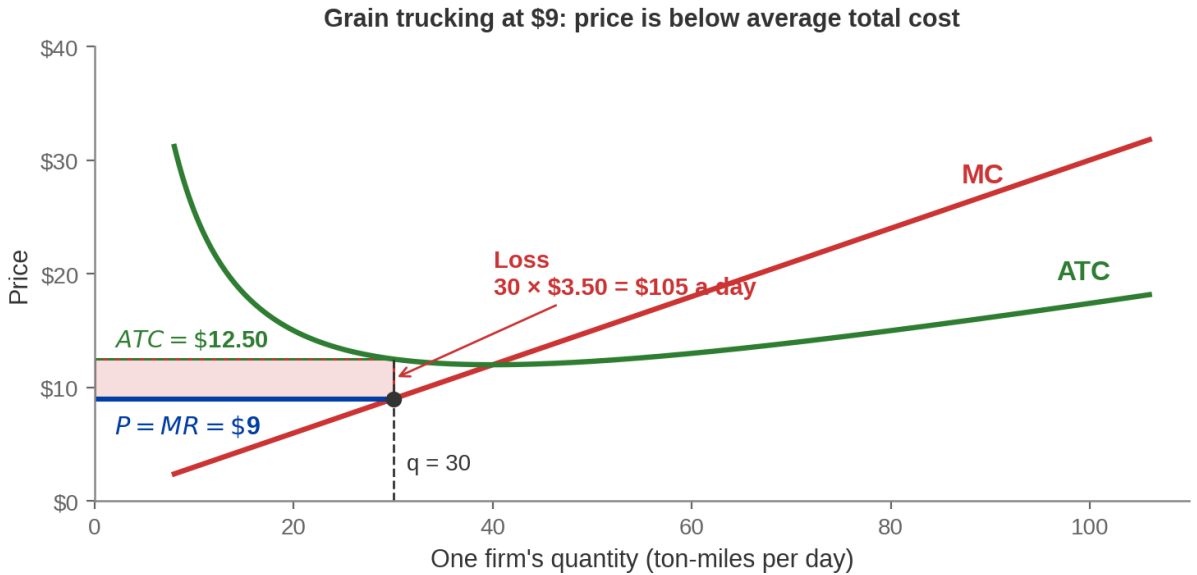
Exhibit 7.10 — Entry, and What It Does to the Price



Note for anyone comparing with the textbook figure: entry adds firms, so market supply shifts RIGHT and the price falls. Some printings of this exhibit draw the shift the other way.

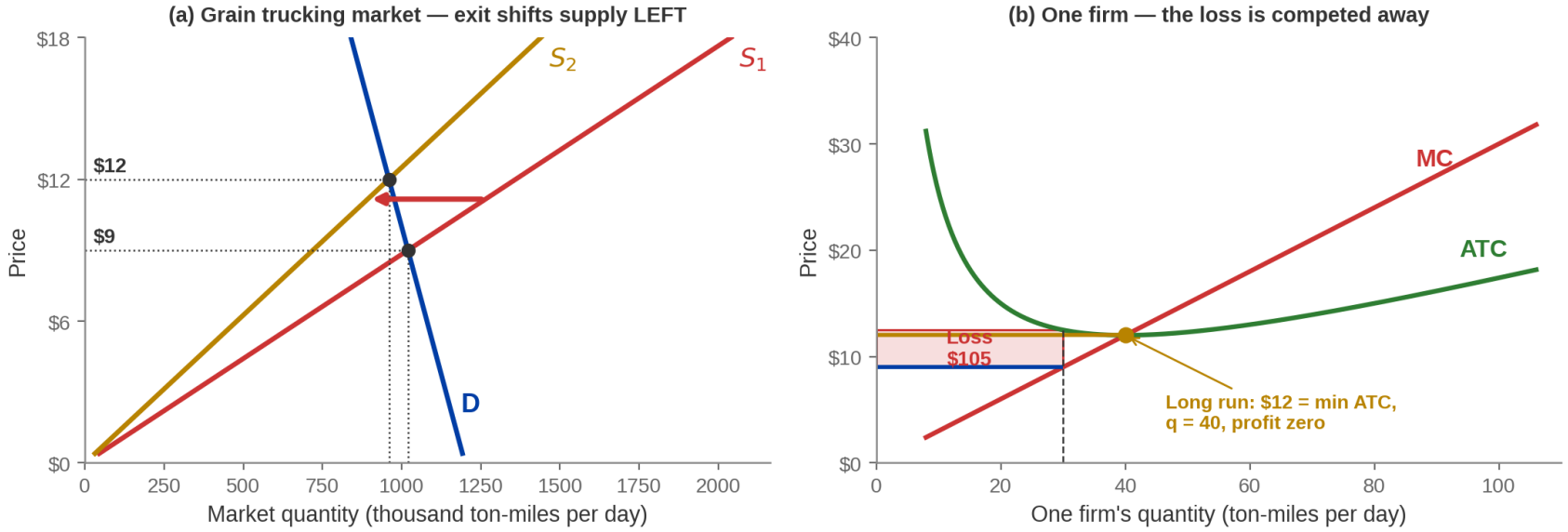
Entry does not stop when the newcomers arrive. It stops when the profit does.

Exhibit 7.11 — And an Industry Losing Money



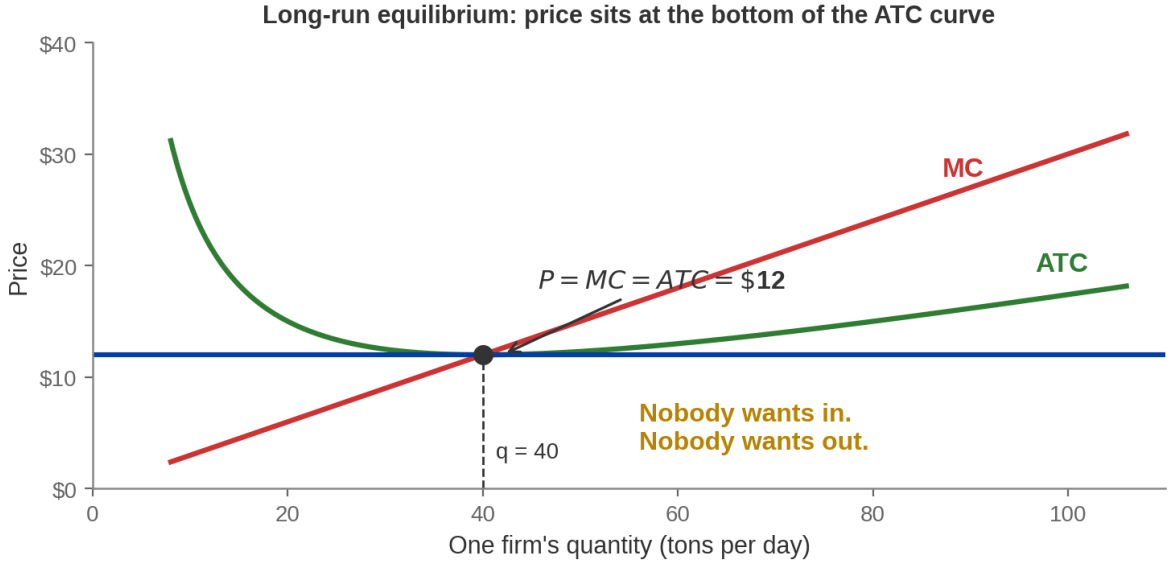
q = 30 ton-miles a day · price $9 · ATC $12.50 · loss $105 a day, per firm

Exhibit 7.12 — Exit, and the Same Mechanism Backwards



Firms leaving is not the market breaking. It is the market moving resources somewhere they are worth more.

Where It Stops



Long-run equilibrium: P = MC = minimum ATC, and economic profit is zero in every industry at once.

Entry and Exit Are the Market Working

PROFITS ATTRACT

A profitable industry is one where resources are earning more than they could elsewhere. Entry is the economy noticing and responding.

LOSSES REPEL

A loss-making industry is one where resources are earning less than they could elsewhere. Exit is the same mechanism running the other way.

Free entry and free exit are what make the reallocation possible. Block either one and the mechanism stops.

Firms closing is uncomfortable. It is also how the economy stops wasting things.

PART 3

Prices Guide the Invisible Hand

A Price Is a Message

A price is the shortest possible summary of everything anyone knows about how scarce something is and how badly it is wanted.

TO BUYERS IT SAYS

Here is what this costs society to give you. Take it only if it is worth that much to you.

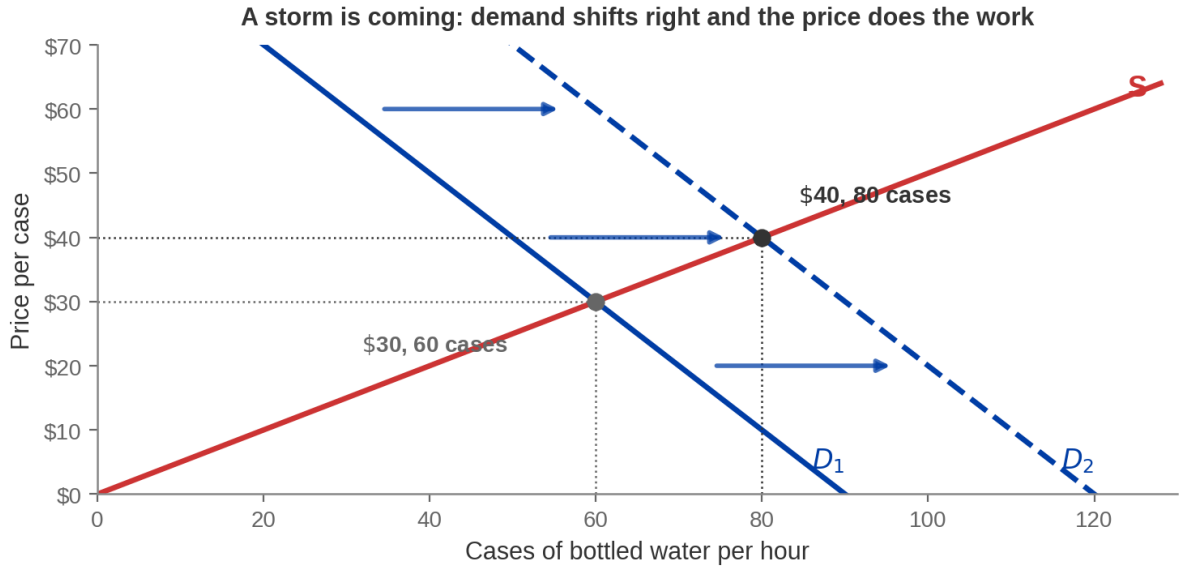
TO SELLERS IT SAYS

Here is what people will give up for this. Make it only if you can do so for less.

TO EVERYONE ELSE IT SAYS

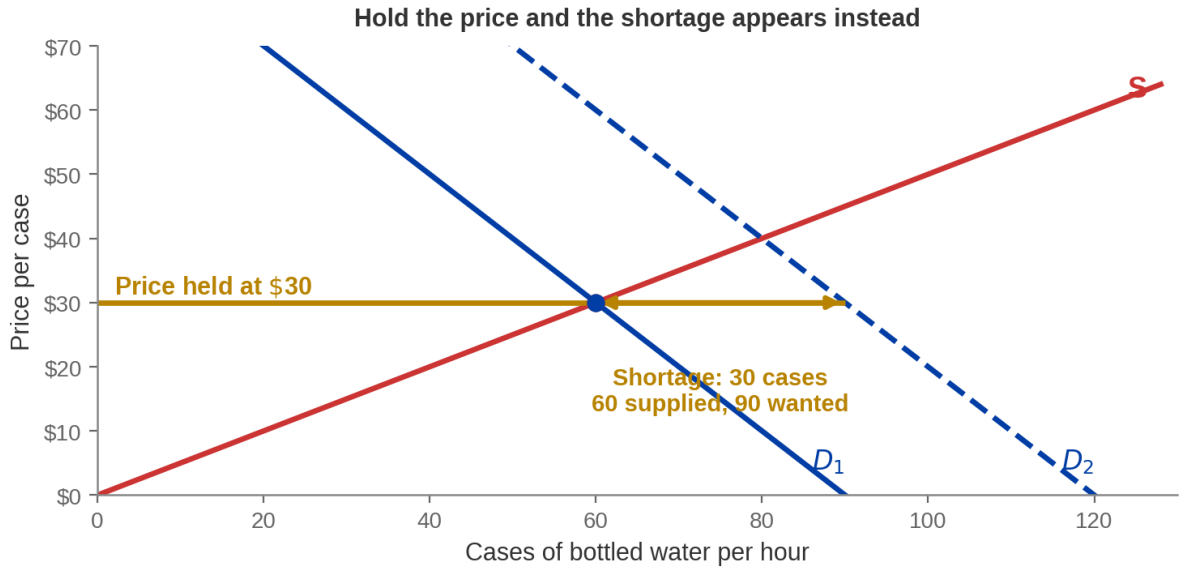
There is money to be made here, or there is not. Come, or go.

Exhibit 7.14 — A Storm Is Coming



Demand shifts right · the price rises from $30 to $40 · quantity traded rises from 60 to 80 cases

Exhibit 7.15 — Hold the Price Instead



At $30: 60 cases arrive, 90 are wanted. The 30-case gap is not filled by anything.

Price Controls Restrict Efficiency



If the price cannot move, it cannot carry information — and everything the invisible hand does, it does through the price.

WHAT IS BLOCKED

Sellers do not learn that supply is short. Buyers do not learn to economize. Nobody outside learns to come and help.

WHAT APPEARS INSTEAD

A shortage, a queue, and an allocation nobody chose and nobody can defend on efficiency grounds.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Deadweight Loss

Deadweight loss — the social surplus that a market intervention destroys. Not transferred to anyone. Destroyed.

A TRANSFER

A price cap moves money from sellers to the buyers who still get the good. Somebody loses, somebody gains, and the total is unchanged.

A DEADWEIGHT LOSS

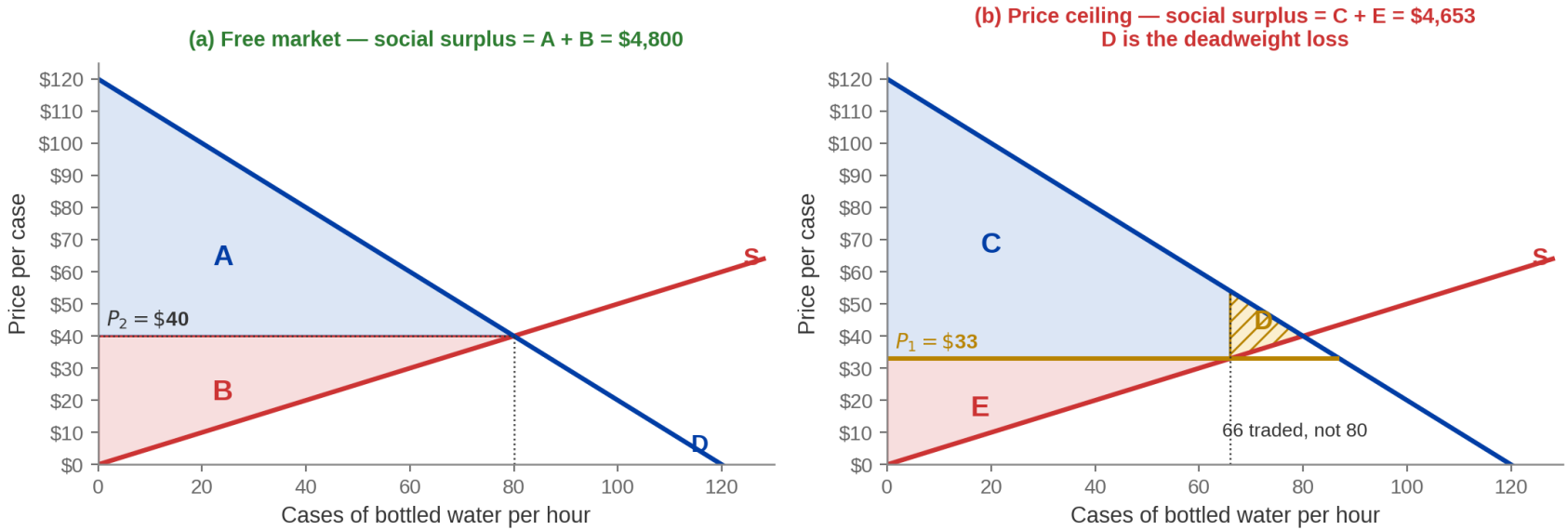
The trades that no longer happen at all. The buyer's gain and the seller's gain both vanish. Nobody picks them up.

YOU ALREADY MET ONE

The $20 in the capped phone market

Ask of any policy: how much is a transfer, and how much simply disappears?

Exhibit 7.16 — Where the Surplus Goes



A + B = $4,800. C + E = $4,653. D = $147, and it belongs to nobody.

The Case on the Other Side

THE ARGUMENT FOR THE CEILING

In a disaster, willingness to pay tracks ability to pay. Letting the price clear the market allocates water to the richest, not the thirstiest. A queue is a worse allocator than a price in normal times and arguably a better one here.

THE ARGUMENT AGAINST

The high price is what brings the extra water in. Cap it and the trucks do not come, so the shortage is deeper and lasts longer — and the people who lose most are the ones without the time to queue.

Two Problems Any Economy Has to Solve

THE COORDINATION PROBLEM

Getting millions of self-interested strangers to form markets at all — to find each other, agree what to trade, and settle on terms. Nobody has a list of who wants what.

THE INCENTIVE PROBLEM

Getting them to actually do the thing once the market exists — to show up, produce, and not shirk, when nobody is watching closely enough to check.

Prices solve both at once. That is why they are the answer to two questions and not just one.

A system that solved only one of these would not work at all.

Two Ways to Answer Them

MARKET ECONOMY

Prices direct the flow of resources and provide the incentives to follow them. Information stays where it already is — with the people who have it — and nobody needs to collect it.

COMMAND ECONOMY

A central agency directs resources and supplies incentives. It must first gather the information that prices would otherwise have summarized, and it must gather it faster than it goes stale.

The Central Planner

Kmart's blue light special: a flashing light, a loudspeaker, and a discount that lasted fifteen minutes.

AS DESIGNED

A store manager, looking at their own overstocked shelves, decided what to discount and when.

AS CENTRALISED

Head office chose the item, the discount and the hour — for every store in the country at once.



Image: Acemoglu, Laibson & List, Microeconomics 3e

PART 4

Equity and Efficiency

Equity

Equity asks whether the distribution of resources across a society is fair.

EFFICIENCY IS POSITIVE

It asks what IS. Given the curves, social surplus is $4,800 or it is not. You can check it.

EQUITY IS NORMATIVE

It asks what OUGHT to be. Nothing in the diagram answers it, and no amount of better data will.

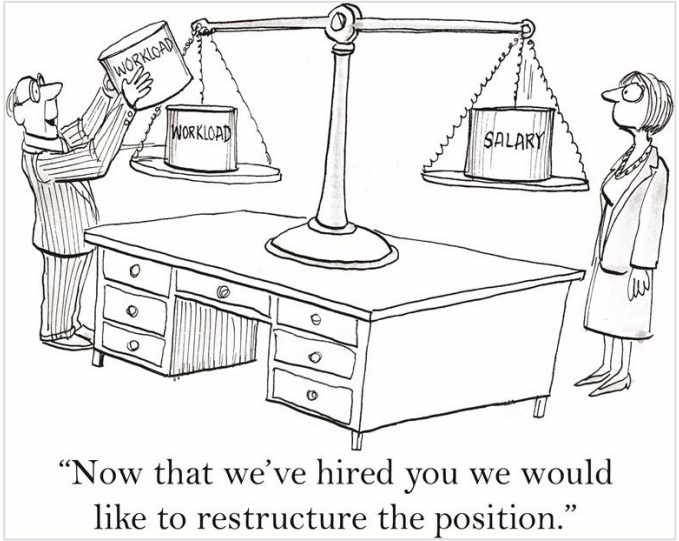


Image: Acemoglu, Laibson & List, Microeconomics 3e

Efficient Is Not the Same as Fair

Every allocation on the frontier is efficient. The frontier does not tell you which point on it to want.

THE TRADE-OFF IS REAL

Most ways of redistributing change behaviour, and changed behaviour costs surplus. Bigger pie, less equal slices — often, not always.

BUT NOT ALWAYS

Some inefficiencies fall hardest on the poor, so removing them raises both. Efficiency and equity are not automatically opposed.

WHAT ECONOMICS CONTRIBUTES

The price of each option. It tells you what the choice costs. It does not tell you which to choose.

One of the roles of government is to address efficient outcomes that we do not consider equitable.

Uber Surge Hour Pricing

It is raining, the game has just ended, and the app says your ride home costs $32 instead of $15.



Is that price doing something useful, or just taking advantage of you?

BOTH SIDES ARE PRICE TAKERS

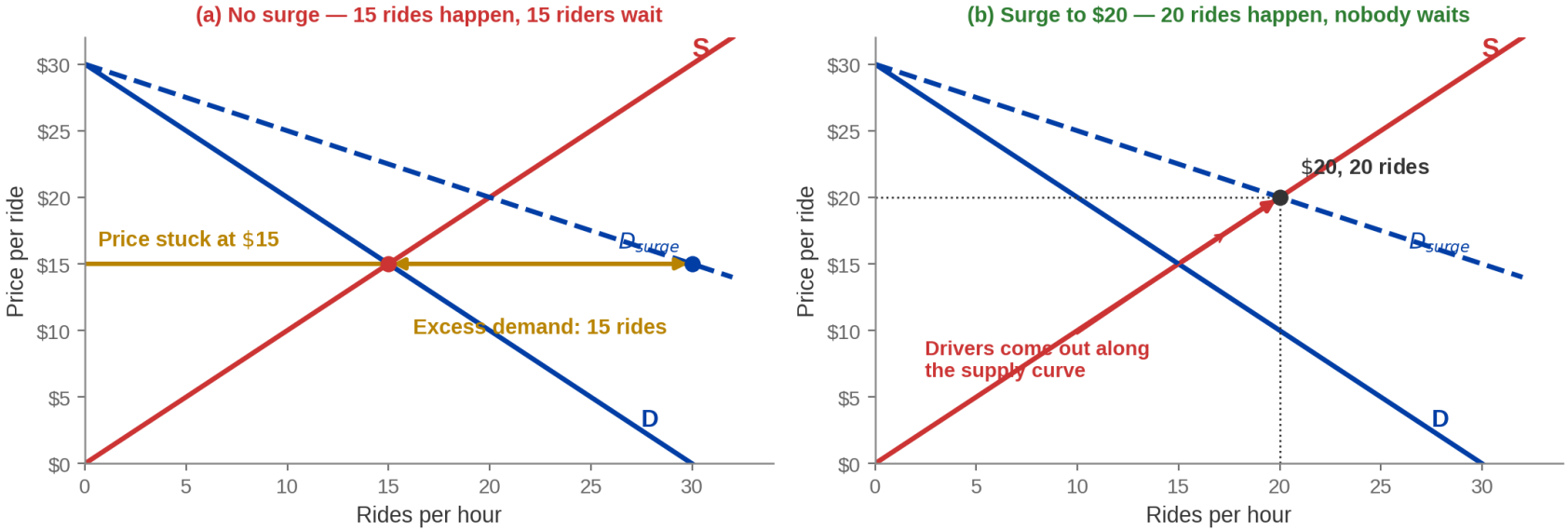
Riders see a price and decide. Drivers see a price and decide. Neither can move it.

THE PRICE IS DOING A JOB

It rations the rides that exist, and it calls more drivers onto the road.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Exhibit 7.17 — Two Ways to Handle a Demand Surge



Without the surge, 15 rides happen and 15 people wait. With it, 20 rides happen and nobody does.

Chapter 7: The Full Picture

	What it says	Key terms	Where you saw it
Efficiency	Competitive equilibrium maximises social surplus.	consumer and producer surplus, Pareto efficiency	Exhibits 7.1–7.3
Across industries	Profit attracts entry, loss forces exit; both stop at P = MC = min ATC.	free entry and exit, long-run equilibrium	Exhibits 7.9–7.12
Prices	Freeze a price and you get a shortage, a queue and a deadweight loss.	price signal, price ceiling, deadweight loss	Exhibits 7.14–7.16
Equity	Efficiency is positive, equity is normative — and the model cannot settle it.	equity, positive and normative	Section 7.5
The evidence	Kerb pricing, a real 10% price cap and surge pricing all show the mechanism.	SFpark, anti-gouging rules, surge pricing	Exhibit 7.17

Three chapters, one argument: buyers optimise, sellers optimise, and the price makes the two add up.

KEY TAKEAWAYS

Chapter 7: Perfect Competition and the Invisible Hand

- Social surplus measures what a market creates. The competitive equilibrium maximises it — restricting trade or forcing it both destroy value.
- Pareto efficient is a low bar. It only means no free moves are left — one person owning everything clears it.
- Profit and loss are signals. Free entry and exit act on them until economic profit is zero and price equals minimum average total cost.
- A price is information and an incentive at once. Freeze it and you get a shortage, a queue, and a deadweight loss nobody receives.
- Efficient is not the same as fair. Economics prices the trade-off; it does not choose for you.